

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 12: Reflection

Step-by-step solutions to the previous year questions

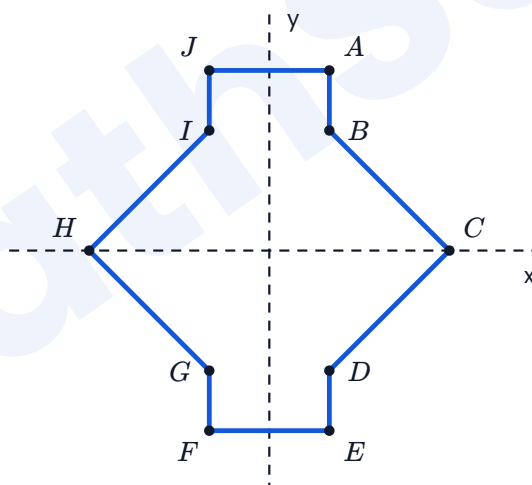
The question numbers match the PYQ question sheet.

1 Reflection of figures

1. Plot $A(1, 3)$, $B(1, 2)$ and $C(3, 0)$ on graph paper. (a) Reflect A and B in the x -axis to get E and D . (b) Reflect A and B in the origin to get F and G . (c) Reflect A , B and C in the y -axis to get J , I and H . (d) Join $A, B, C, D, E, F, G, H, I, J$ in order and name the closed figure. **ICSE 2025, 5 marks**

SOLUTION

- (a) Reflection in the x -axis changes the sign of y : $E(1, -3)$ and $D(1, -2)$.
 (b) Reflection in the origin changes both signs: $F(-1, -3)$ and $G(-1, -2)$.
 (c) Reflection in the y -axis changes the sign of x : $J(-1, 3)$, $I(-1, 2)$ and $H(-3, 0)$.



- (d) The figure has 10 sides and two of its angles point inwards on each side.

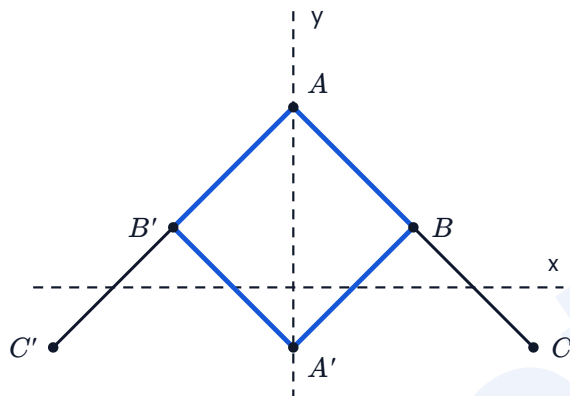
Answer: a concave decagon, shaped like a double-headed arrow

2. Plot $A(0, 3)$, $B(2, 1)$ and $C(4, -1)$. (a) Reflect B and C in the y -axis to get B' and C' , and write their coordinates. (b) Reflect A in the line BB' to get A' , and write its coordinates. (c) Join A, B, A', B' in order and name the figure. **ICSE 2024, 5 marks**

SOLUTION

(a) Reflection in the y -axis changes the sign of x : $B'(-2, 1)$ and $C'(-4, -1)$.

(b) BB' is the horizontal line $y = 1$. A is 2 units above it, so A' is 2 units below it: $A'(0, -1)$.



(c) $AB = BA' = A'B' = B'A = \sqrt{2^2 + 2^2} = \sqrt{8}$, and the diagonals $AA' = BB' = 4$ are equal.

Answer: $B'(-2, 1)$, $C'(-4, -1)$, $A'(0, -1)$; $ABA'B'$ is a square

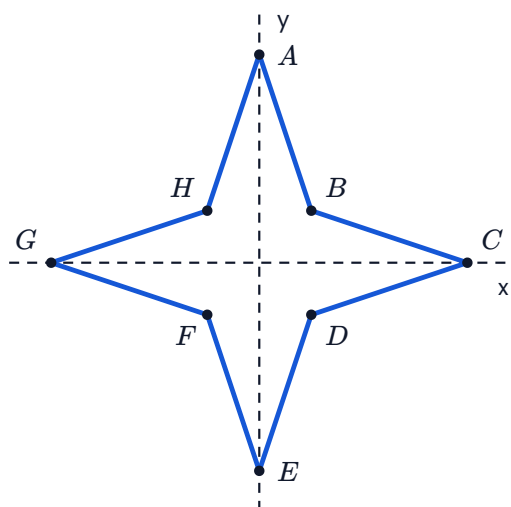
3. Plot $A(0, 4)$, $B(1, 1)$ and $C(4, 0)$. (a) Reflect A and B in the x -axis to get E and D . (b) Reflect B in the origin to get F , and write its coordinates. (c) Reflect B and C in the y -axis to get H and G . (d) Join $A, B, C, D, E, F, G, H, A$ in order and name the closed figure. **ICSE 2023, 5 marks**

SOLUTION

(a) $E(0, -4)$ and $D(1, -1)$.

(b) Reflection in the origin changes both signs: $F(-1, -1)$.

(c) $H(-1, 1)$ and $G(-4, 0)$.



(d) The figure has 8 sides, with four of its corners pointing inwards.

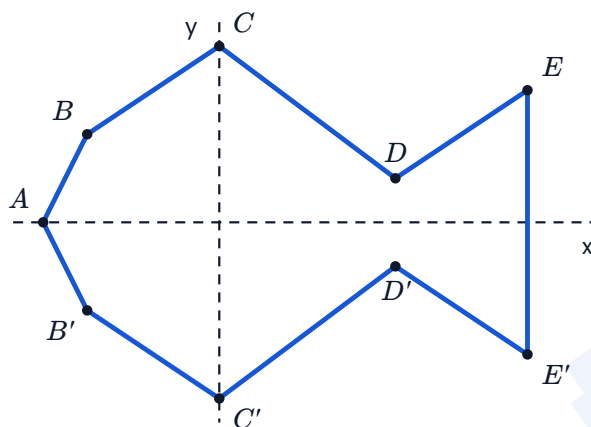
Answer: $F(-1, -1)$; the figure is a concave octagon in the shape of a four-pointed star

4. Plot $A(-4, 0)$, $B(-3, 2)$, $C(0, 4)$, $D(4, 1)$ and $E(7, 3)$. Reflect B , C , D and E in the x -axis to get B' , C' , D' and E' . Join $A, B, C, D, E, E', D', C', B', A$ in order and name the closed figure.

ICSE 2020, 4 marks

SOLUTION

Reflection in the x -axis changes the sign of y : $B'(-3, -2)$, $C'(0, -4)$, $D'(4, -1)$ and $E'(7, -3)$.



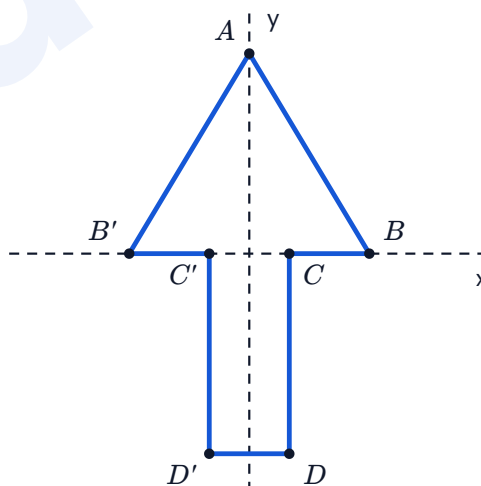
Answer: the figure is shaped like a fish (a concave nonagon, symmetric about the x -axis)

5. Plot $A(0, 5)$, $B(3, 0)$, $C(1, 0)$ and $D(1, -5)$. Reflect B , C and D in the y -axis to get B' , C' and D' , and write their coordinates. Join $A, B, C, D, D', C', B', A$ in order and name the closed figure.

ICSE 2019, 4 marks

SOLUTION

Reflection in the y -axis changes the sign of x : $B'(-3, 0)$, $C'(-1, 0)$ and $D'(-1, -5)$.

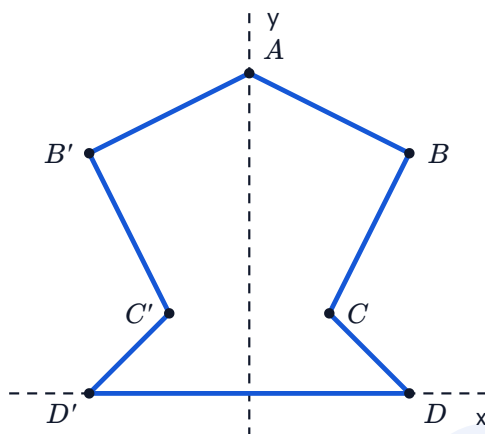


Answer: $B'(-3, 0)$, $C'(-1, 0)$, $D'(-1, -5)$; the figure is an arrow head (a concave heptagon)

6. Plot $A(0, 4)$, $B(2, 3)$, $C(1, 1)$ and $D(2, 0)$. Reflect B , C and D in the y -axis to get B' , C' and D' , and write their coordinates. Join $A, B, C, D, D', C', B', A$ in order and write the equation of the line of symmetry of the figure. **ICSE 2017, 5 marks**

SOLUTION

Reflection in the y -axis changes the sign of x : $B'(-2, 3)$, $C'(-1, 1)$ and $D'(-2, 0)$.



The figure is its own mirror image in the y -axis.

Answer: $B'(-2, 3)$, $C'(-1, 1)$, $D'(-2, 0)$; the line of symmetry is $x = 0$

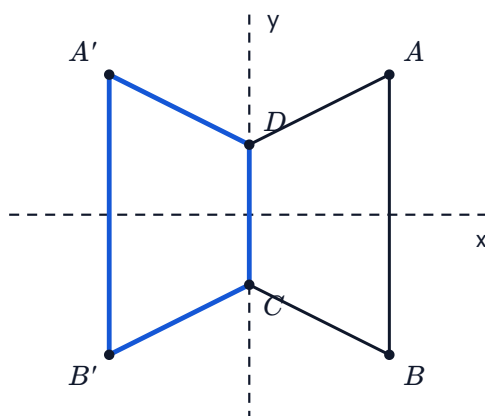
2 Invariant points

7. $ABCD$ is a quadrilateral with vertices $A(2, 2)$, $B(2, -2)$, $C(0, -1)$ and $D(0, 1)$. (a) Reflect $ABCD$ in the y -axis to get $A'B'CD$, and write the coordinates of A' and B' . (b) Name two points that are invariant under this reflection. (c) Name the polygon $A'B'CD$. **ICSE 2018, 4 marks**

SOLUTION

(a) $A'(-2, 2)$ and $B'(-2, -2)$.

(b) C and D lie on the y -axis, which is the mirror line, so they are invariant.



(c) $A'B' \parallel DC$ (both vertical) and $A'D = B'C = \sqrt{5}$.

Answer: $A'(-2, 2)$, $B'(-2, -2)$; invariant points C and D ; $A'B'CD$ is an isosceles trapezium

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